PCCST501 – Computer Networks

Assignment 3: LAN, Ethernet & Wireless Network Investigation

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Q1)Explain the responsibilities of the data-link layer and distinguish Logical Link Control (LLC) from Media Access Control (MAC).

Data-link layer:

The Data-Link Layer is responsible for communication between two adjacent nodes over a link. While the Network Layer provides host-to-host communication, the Data-Link Layer provides node-to-node delivery across each individual link.

As explained in Section 5.1 of Datalink Layer 1.pdf, the Data-Link Layer can be divided into two sublayers: Data Link Control (DLC) and Media Access Control (MAC).

Responsibilities of the data-link layer:

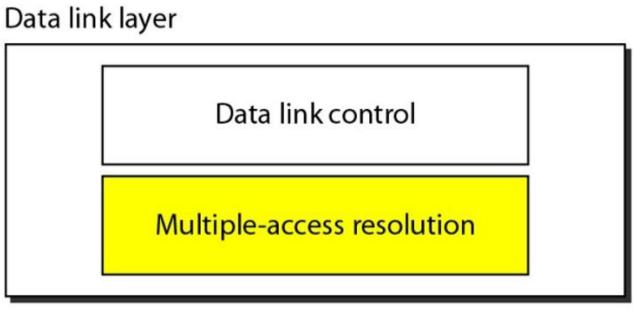
1. Framing: The Data-Link Layer divides the bit stream into units called frames. Each frame can be identified and processed separately.
2. Physical addressing: It uses link-layer addresses to identify the source and destination interfaces on a local link.
3. Flow control: It coordinates the rate at which frames are sent so that the receiver can process them.
4. Error control: It can detect damaged or lost frames and, where the protocol supports it, coordinate retransmission.
5. Error detection and correction: Redundant information is used to detect transmission errors. Some protocols also support error correction.
6. Media access control: On a shared or broadcast link, the MAC sublayer determines how stations access the transmission medium.

Difference between LLC and MAC:

The assignment asks for LLC and MAC. LLC stands for Logical Link Control, while MAC stands for Media Access Control. The notes explain the related division using the terms DLC and MAC: DLC covers common link-control functions, while MAC handles issues specific to broadcast links.

Feature	LLC (Logical Link Control)	MAC (Media Access Control)
Position	Upper sublayer of the Data-Link Layer	Lower sublayer of the Data-Link Layer
Main purpose	Provides the logical interface between the Network Layer and link-layer services	Controls access to the transmission medium
Main functions	Supports communication with upper-layer protocols and logical link services	Handles MAC addressing, frame transmission and medium access

Scope	Logical link services	Access to a shared or broadcast medium
Example	LLC service interface	Ethernet MAC and IEEE 802.11 MAC



Upper sublayer – responsible for data link control
- **Called LLC – for flow and error control**

Lower sublayer - responsible for resolving access the shared media
- **Called MAC – for multiple acceess resolution**

Conclusion: The Data-Link Layer provides node-to-node communication through framing, addressing, flow control and error handling. LLC provides the logical link interface, whereas MAC manages access to the transmission medium and link-layer addressing.

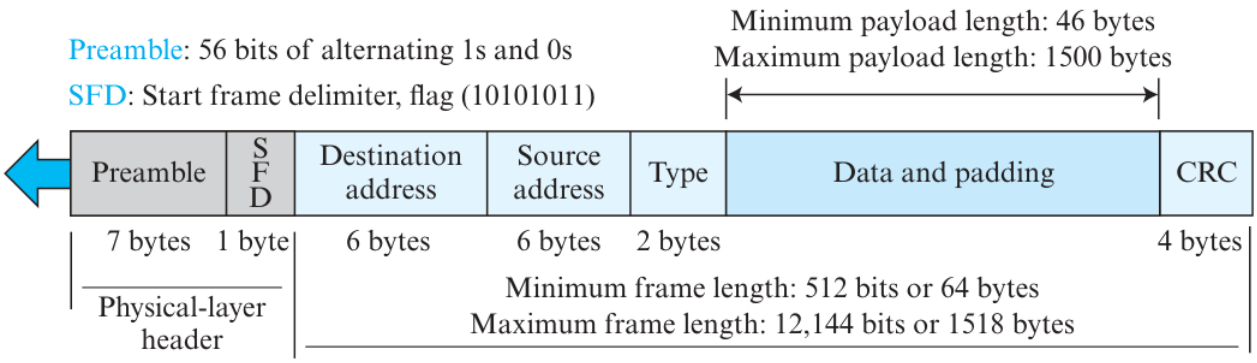
Q2) Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

Ethernet frame format:

Ethernet is a widely used LAN technology. An Ethernet frame carries data from an upper-layer protocol and includes addressing, protocol-identification and error-detection fields.

Ethernet frame diagram:

Figure 5.55 *Ethernet frame*



Field	Size	Purpose
Preamble	7 bytes	Contains alternating bits to alert the receiver and help synchronize timing.
Start Frame Delimiter (SFD)	1 byte	Indicates the beginning of the Ethernet frame and identifies the start of the destination address field.
Destination MAC address	6 bytes	Identifies the intended destination interface or group of interfaces.
Source MAC address	6 bytes	Identifies the transmitting interface.
Type	2 bytes	Identifies the upper-layer protocol carried in the frame, such as IP or ARP, in Ethernet II framing.
Data and padding	46–1500 bytes	Carries the upper-layer packet. Padding is added when necessary to reach the minimum frame size.
CRC / FCS	4 bytes	Contains a CRC-32 value used by the receiver to detect frame corruption.

Conclusion: Ethernet frames provide source and destination addressing, identify the encapsulated protocol and carry error-detection information. The FCS enables the receiver to detect many transmission errors.

Q3) Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.

Aim:

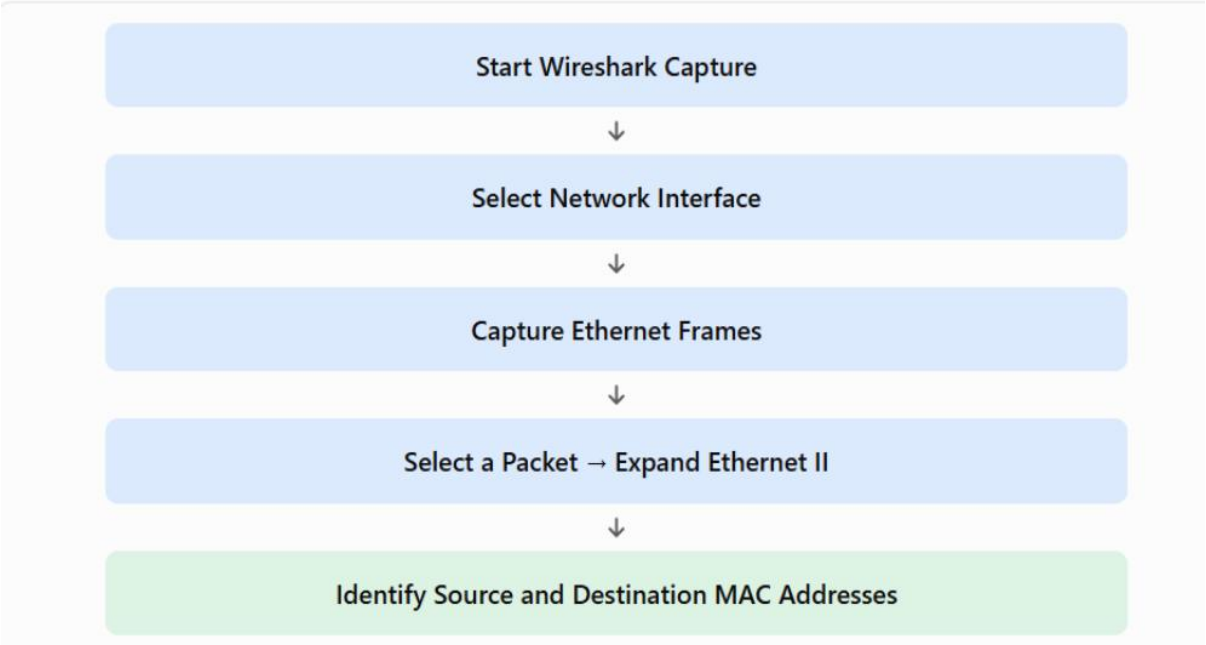
To capture Ethernet traffic using Wireshark and identify the source and destination MAC addresses of at least five captured frames.

Requirements:

- A computer connected to a network.
- Wireshark packet-analysis software.
- An active Ethernet network interface.
- Network traffic generated by normal communication or a command such as ping.

Procedure:

1. Open Wireshark.
2. Select the active Ethernet interface from the available interfaces.
3. Start packet capture.
4. Generate network traffic by opening a website or using ping to contact a reachable host.
5. Allow sufficient packets to be captured, then stop the capture.
6. Apply the display filter eth to help inspect Ethernet-decoded packets.
7. Select a packet from the packet list.
8. Expand the Ethernet II section in the packet details pane.
9. Identify the source MAC address, destination MAC address and Type field.
10. Repeat the procedure for at least five frames.
11. Record the observed values in the table below.
12. Save the capture file and take a readable screenshot showing the packet list and Ethernet details.



Observation table:

Frame number	Source MAC address	Destination MAC address
1	A0:B1:C2:D3:E4:F5	FF:FF:FF:FF:FF:FF
2	12:34:56:78:9A:BC	A0:B1:C2:D3:E4:F5
3	A0:B1:C2:D3:E4:F5	12:34:56:78:9A:BC

4	12:34:56:78:9A:BC	A0:B1:C2:D3:E4:F5
5	A0:B1:C2:D3:E4:F5	12:34:56:78:9A:BC

Interpretation

- The source MAC address identifies the interface that transmitted the frame on the observed link.
- The destination MAC address identifies the intended receiver or destination group on that link.
- The Type field identifies the upper-layer protocol in Ethernet II framing.
- The addresses observed depend on the packet selected. For example, a frame carrying an ARP request can have a broadcast destination MAC address.

Result :Ethernet traffic was captured using Wireshark. The source and destination MAC addresses of five frames can be identified from the Ethernet II header in the packet details pane

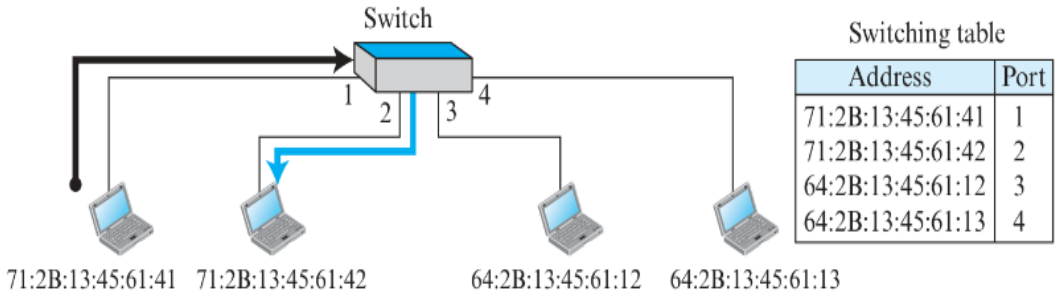
Q4) Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.

Introduction:

A link-layer switch operates at the Physical and Data-Link Layers. It regenerates the received signal and examines the source and destination MAC addresses contained in Ethernet frames.

Unlike a hub, a switch has filtering capability and uses a switching table to decide through which port a frame should be forwarded.

Figure 5.85 *Link-Layer Switch*



Operation of a switch:

1. Frame reception: The switch receives an Ethernet frame through one of its ports.
2. MAC address learning: It examines the source MAC address and associates that address with the incoming port.

3. Table lookup: The switch checks the destination MAC address against its forwarding table.
4. Known destination: If the destination MAC address is known, the switch forwards the frame through the corresponding port.
5. Unknown destination: If the destination is not known, the switch floods the frame through eligible ports other than the incoming port.
6. Table updating: As new frames arrive, the switch learns additional source addresses and updates the table when devices move between ports. Learned entries are aged out if not refreshed.

Example of MAC address learning:

Assume Host A is connected to Port 1, Host B to Port 2, and Host C to Port 3.

Step	Frame received	Learning action	Forwarding action
1	A → unknown destination	Learn A → Port 1	Flood to eligible ports except Port 1
2	B → A	Learn B → Port 2	Forward to Port 1
3	A → B	Refresh A → Port 1	Forward to Port 2
4	C → A	Learn C → Port 3	Forward to Port 1

After these frames, the switch's learned forwarding table contains:

MAC address	Port
A	1
B	2
C	3

This example illustrates the learning and filtering process described in Figure 5.85

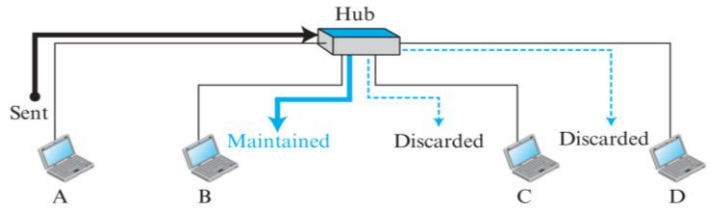
Conclusion: A switch learns source MAC addresses and their incoming ports. It uses the forwarding table to send known unicast frames through the correct port and floods frames when the destination is unknown.

Q5) Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.

Hub:

A hub is a Physical Layer device. It receives a signal, regenerates it and sends it through all outgoing ports except the incoming port. It does not examine MAC addresses and cannot filter frames.

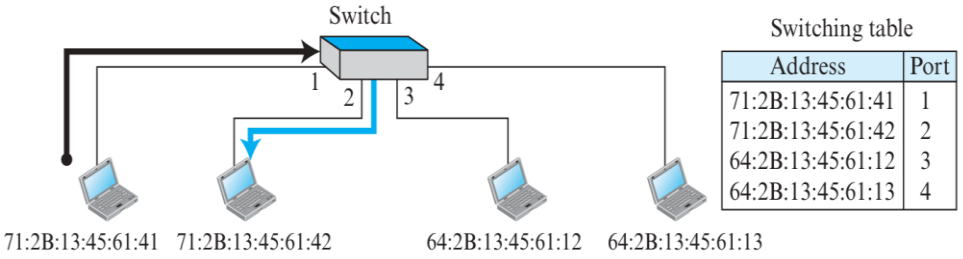
Figure 5.84 *Repeater or hub*



Switch:

A link-layer switch operates at the Physical and Data-Link Layers. It examines MAC addresses and uses a forwarding table to select the appropriate outgoing port. This reduces unnecessary frame transmission compared with a hub.

Figure 5.85 *Link-Layer Switch*



Switching table

Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3
64:2B:13:45:61:13	4

Bridge:

A bridge is a Data-Link Layer device used to connect LAN segments. It learns MAC addresses and filters or forwards frames between the connected segments. A bridge and a switch perform similar Layer 2 functions, although modern switches generally provide more ports and higher forwarding capacity.

Comparison table:

Feature	Hub	Bridge	Switch
OSI layer	Layer 1 – Physical	Layer 2 – Data-Link	Layer 2 – Data-Link, with Physical Layer functions
Forwarding method	Repeats signals through outgoing ports	Uses MAC addresses to forward or filter frames between segments	Uses a MAC forwarding table to select the outgoing port
MAC address learning	No	Yes	Yes
Filtering capability	No	Yes	Yes

Collision domains	One shared collision domain	Separates collision domains between connected segments	Each port is a separate collision domain
Broadcast domains	Does not separate broadcast traffic	Normally one shared broadcast domain	Normally one broadcast domain per VLAN
Practical use	Legacy shared Ethernet	Connecting and filtering LAN segments	Modern LAN connectivity

Conclusion: A hub repeats signals without learning addresses. Bridges and switches use MAC addresses to filter and forward frames. Switches are preferred in modern LANs because they support efficient per-port forwarding.

Q6) Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.

Definition:

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It is a random-access method used in shared, half-duplex Ethernet networks. It allows stations to sense the medium, transmit when appropriate, detect collisions and retry transmission after a collision.

Working of CSMA/CD:

1. Carrier sensing: A station listens to the medium before transmitting.
2. Transmission: If the medium is idle, the station transmits its frame. If the medium is busy, it waits according to the applicable persistence procedure.
3. Collision detection: The transmitting station continues monitoring the medium while sending.
4. Collision handling: If a collision is detected, the station aborts the transmission and sends a jam signal.
5. Backoff: The station waits for a random interval determined by the binary exponential backoff procedure.
6. Retransmission: After the waiting interval, the station attempts to transmit again.

Why CSMA/CD is less important in modern switched Ethernet:

Traditional shared Ethernet allows multiple stations to compete for the same transmission medium. Collisions can occur when two stations transmit at the same time.

Modern switched Ethernet generally uses dedicated point-to-point links operating in full-duplex mode. Each device can transmit and receive simultaneously, so the

shared-medium collisions addressed by CSMA/CD do not occur during normal full-duplex operation.

Therefore:

- A switch provides separate links to connected devices.
- Full-duplex operation eliminates the need for collision detection and collision-based backoff.
- Network performance improves because devices do not compete for a single shared collision medium.
- CSMA/CD remains important for understanding legacy shared Ethernet and half-duplex Ethernet.

Conclusion: CSMA/CD detects and handles collisions in shared half-duplex Ethernet. Its importance has decreased because modern switched full-duplex Ethernet uses dedicated links and does not normally experience such collisions.

Q7) Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.

Definition:

CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is the access method used by the Distributed Coordination Function (DCF) of IEEE 802.11 wireless LANs.

The wireless channel is shared by multiple stations. Because a wireless station may not be able to detect a collision while transmitting, CSMA/CA attempts to reduce the probability of collisions before transmission.

Why CSMA/CD is unsuitable for wireless LANs:

The notes explain three important difficulties:

1. A wireless station generally cannot transmit and receive simultaneously in the manner required for collision detection.
2. The hidden-station problem can cause two stations to transmit to the same receiver without detecting one another.
3. Distance and signal fading can prevent a station from detecting a collision occurring elsewhere in the network.

Figure 6.3 *Hidden station problem*

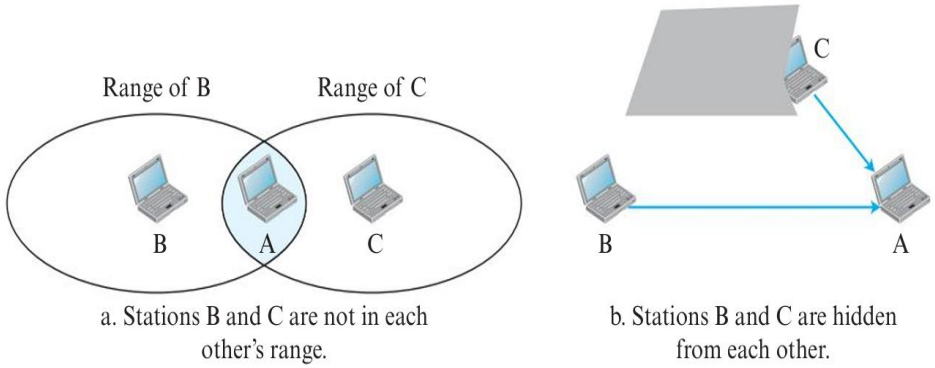


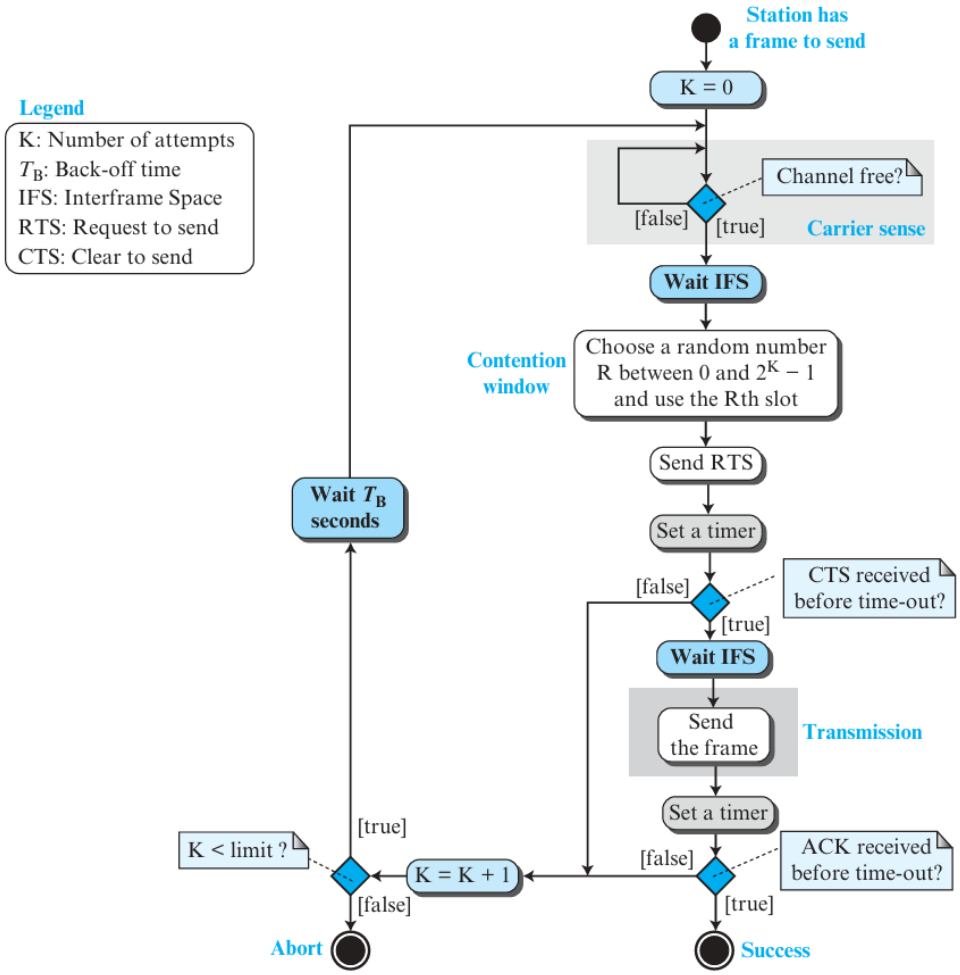
Figure 6.3, Hidden Station Problem, illustrates the second difficulty. Stations B and C may be unable to hear one another even though both can reach Station A. If both transmit to A at the same time, their frames can collide at A.

Working of CSMA/CA:

1. Carrier sensing: A station checks whether the wireless channel is busy or idle.
2. Interframe Space (IFS): If the channel is idle, the station waits for the required interframe interval.
3. Contention window: The station chooses a random backoff interval from the contention window.
4. Channel monitoring: The backoff counter decreases while the channel is idle. If the channel becomes busy, the counter is frozen and resumes when the channel is idle again and the required waiting conditions are met.
5. Transmission: When the counter reaches zero, the station transmits the frame.
6. Acknowledgement: The receiver sends an acknowledgement for a successfully received unicast frame.
7. Retransmission: If the acknowledgement is not received, the sender may retry using the applicable backoff procedure.

CSMA/CA flow diagram:

Figure 6.7 *Flow diagram of CSMA/CA*



Comparison of CSMA/CD and CSMA/CA:

Feature	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Main objective	Detect collisions and handle them	Reduce the likelihood of collisions
Application	Legacy shared, half-duplex Ethernet	IEEE 802.11 wireless LAN
Carrier sensing	Yes	Yes
Collision handling	Detects a collision during transmission, aborts and backs off	Uses waiting intervals and random backoff to reduce collisions
Acknowledgement	Not normally used by classic Ethernet MAC to acknowledge every frame	Commonly used for unicast data-frame delivery
Hidden-station problem	Not a characteristic problem of the conventional wired shared medium	Can occur; RTS/CTS may help reduce its effects

Conclusion: CSMA/CD reacts to detected collisions, whereas CSMA/CA attempts to avoid collisions before transmission. CSMA/CA is used in wireless LANs because wireless stations cannot reliably use the collision-detection method of traditional Ethernet.

Q8) Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.

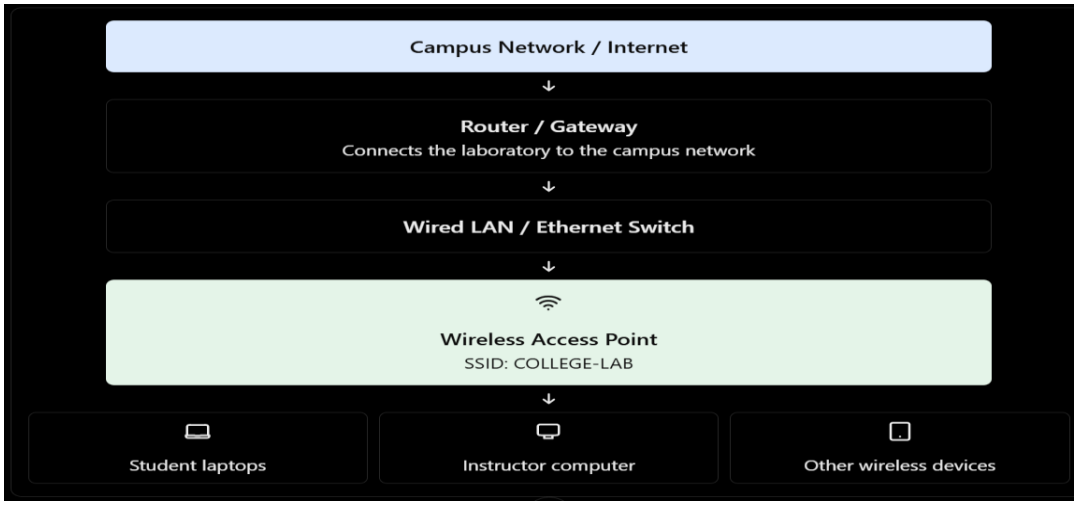
Aim:

To design a wireless LAN that allows students and an instructor to connect their devices to the college network and access permitted network resources.

Network architecture:

The wireless LAN should use an infrastructure architecture. In an infrastructure BSS, wireless stations communicate through an Access Point (AP). The AP connects the wireless network to the wired infrastructure.

Proposed WLAN diagram:



Proposed configuration:

Parameter	Proposed setting	Explanation
WLAN architecture	Infrastructure BSS	Stations connect through an AP rather than forming an independent ad hoc network.
Access point placement	Near the centre of the laboratory, preferably elevated and unobstructed	Provides a practical starting point for distributing the wireless signal. Actual placement

		should be checked by testing coverage.
SSID	COLLEGE-LAB	Identifies the laboratory wireless network.
Frequency/channel	Select an available channel appropriate to the AP, client devices and local radio environment	Reduces interference from nearby networks. Channel selection should be based on a local survey.
IP addressing	Use a subnet assigned by the college network administrator	Prevents address conflicts with existing networks.
Address assignment	DHCP, if provided by the campus network	Allows client devices to obtain their IP configuration automatically.
Security	Use the strongest authentication and encryption supported by the college equipment; prefer institution-managed credentials	Reduces unauthorized network access.
Access control	Restrict access to the services required for laboratory work	Helps protect campus resources.

Channel considerations:

Wireless stations share a radio medium. Nearby networks may interfere with one another, and the signal may weaken because of obstacles and distance.

The following should be considered:

1. Select a channel supported by the AP and client devices.
2. Check nearby networks and interference before choosing a channel.
3. Place the AP where its signal can reach the student workstations.
4. Avoid unnecessary obstructions between the AP and connected devices.
5. If one AP cannot provide sufficient coverage or capacity, consider multiple BSSs connected through a distribution system, forming an ESS

Basic security requirements:

- Configure wireless authentication and encryption.
- Use institution-approved credentials.
- Change default administrative credentials.
- Limit access to network management functions.
- Keep the AP configuration and software maintained.
- Restrict access to sensitive campus resources according to the college's policies.

Conclusion: An infrastructure WLAN with an appropriately placed access point provides wireless connectivity for a college laboratory. Suitable channel selection, organized IP addressing and access control help make the network usable and secure.

Q9) Explain Mobile IP and the need for mobility support when a device moves between networks.

Definition of Mobile IP:

Mobile IP is a network-layer mechanism that allows a mobile host to move from its home network to a foreign network while retaining its permanent home IP address.

When a host moves to another IP network, its network-layer address may need to change. If its address changes during an active communication session, the communication may be interrupted. Mobile IP addresses this problem by associating the mobile host with both a permanent home address and a temporary care-of address.

Components of Mobile IP:

Component	Function
Mobile host	The device that moves from its home network to another network.
Home address	The permanent IP address associated with the mobile host's home network.
Care-of address	A temporary address associated with the mobile host's current location on a foreign network.
Home agent	A router on the home network that acts on behalf of the mobile host and forwards packets toward its current location.
Foreign agent	A router on the foreign network that receives packets forwarded by the home agent and delivers them to the mobile host.
Remote host	The device communicating with the mobile host.

Why are two addresses required?

- The home address remains associated with the mobile host's home network.
- The care-of address identifies its current location on a foreign network.
- The home agent uses the current location information to forward packets to the mobile host.

If a host simply changed its IP address every time it moved, other hosts would need to learn its new address, and active communication could be interrupted. The two-address approach helps avoid this problem.

Working of Mobile IP:

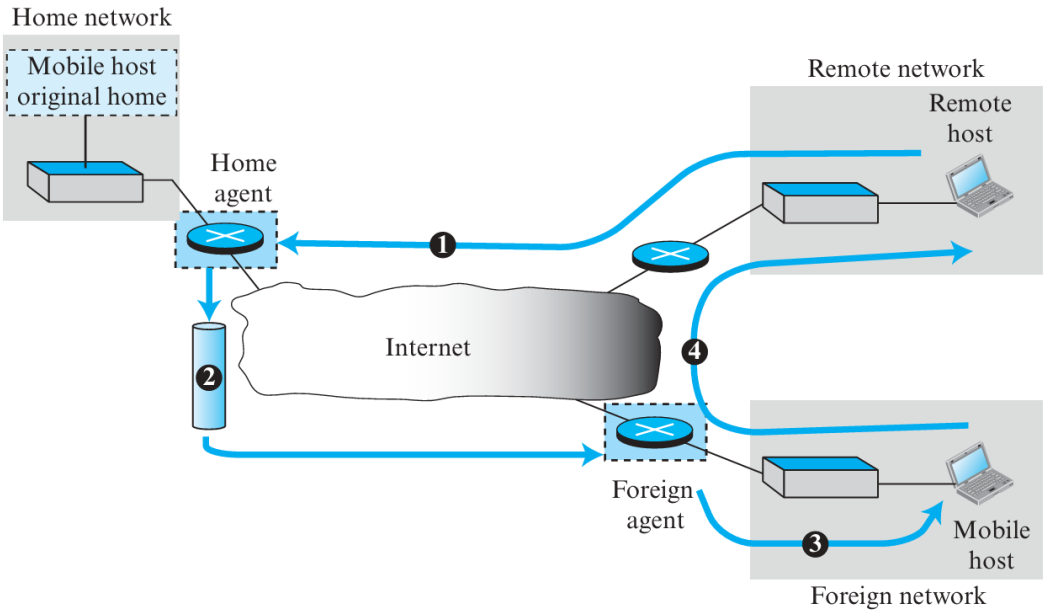
The notes describe three phases: agent discovery, registration and data transfer.

1. Agent discovery: The mobile host discovers the home agent and, after moving, the foreign agent or an available care-of address.
2. Registration: The mobile host registers its current care-of address with the home agent, through the applicable agent arrangement.
3. Data transfer: The home agent intercepts packets addressed to the mobile host's home address and forwards them toward its current location using tunnelling.

During data transfer, the home agent encapsulates the original IP packet inside another IP packet addressed toward the foreign agent. The foreign agent removes the outer encapsulation and delivers the original packet to the mobile host.

Mobile IP diagram:

Figure 6.58 *Data transfer*



Need for mobility support:

Mobile IP provides the following benefits:

1. Continuity of communication: Helps maintain communication while the host is away from its home network.
2. Stable addressing: The home address remains associated with the mobile host.
3. Location management: The home agent maintains information needed to forward packets toward the host's current location.
4. Transparent packet forwarding: The remote host can continue sending packets to the home address.
5. Support for movement between networks: The mobile host can attach to a foreign network and register a care-of address.

Mobile IP does not guarantee that every active connection will remain uninterrupted during every handover; it provides a mechanism to support reachability while the host moves.

Conclusion: Mobile IP separates the mobile host's permanent identity on its home network from its current network location. Through agent discovery, registration and tunnelling, packets can be forwarded to the host after it moves to a foreign network.

Q10)Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Aim: To compare the frame structure of wired Ethernet (IEEE 802.3) and wireless LAN (IEEE 802.11) using packet captures in Wireshark.

Procedure:

1. Open Wireshark and start capturing packets on a wired Ethernet interface.

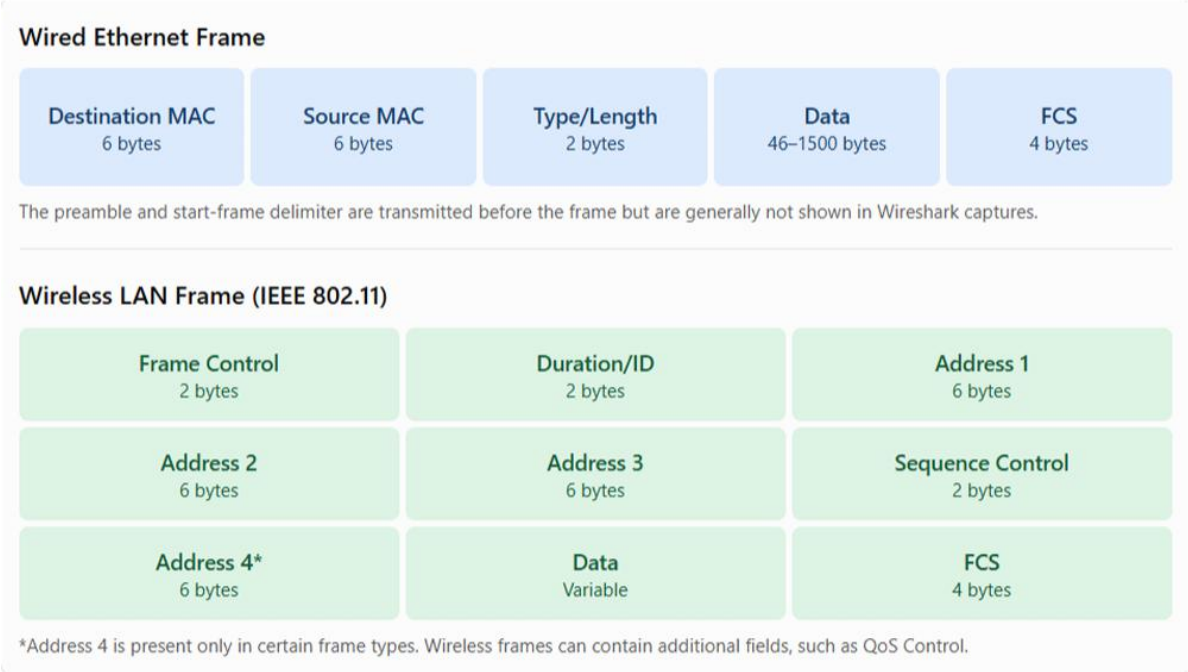
2. Generate network traffic by opening a website or using ping.

3. Apply the filter eth to view Ethernet frames.

4. Capture packets on a Wi-Fi interface and apply the filter wlan.

5. Select packets and expand the Ethernet II and IEEE 802.11 headers to compare their fields.

Frame Format Diagrams:



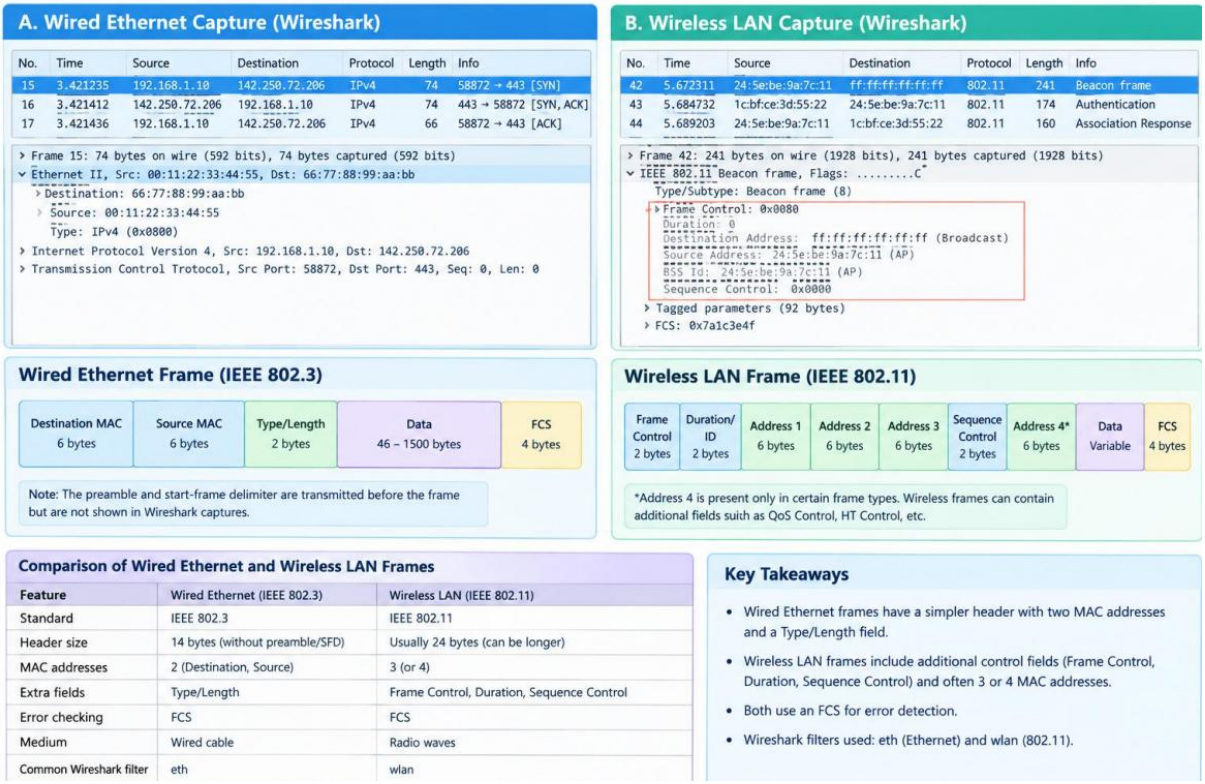
Feature	Wired Ethernet (IEEE 802.3)	Wireless LAN (IEEE 802.11)
Transmission medium	Twisted-pair cable or other wired medium	Radio waves
Frame format	Ethernet frame	802.11 wireless frame
Address fields	Normally contains destination and source MAC addresses	Can contain up to four MAC address fields, depending on the frame type and transmission direction
Header information	Includes destination address, source address, and type/length field	Includes frame control, duration/ID, address fields, and sequence control
Error detection	Uses a Frame Check Sequence (FCS)	Uses an FCS for frame error detection
Access method	Traditional shared Ethernet uses CSMA/CD; modern switched full-duplex Ethernet does not normally use collision detection	Uses CSMA/CA to reduce the likelihood of collisions
Mobility	Devices are generally connected through physical cables	Devices can move within wireless coverage
Wireshark identification	Commonly displayed as Ethernet II	Commonly displayed as IEEE 802.11 when wireless headers are available

Procedure:

1. Open Wireshark and capture packets from a wired Ethernet interface.
2. Select an Ethernet packet and expand its Ethernet header to examine the source MAC address, destination MAC address, type field, and other available fields.
3. Capture packets from a wireless interface in monitor mode, if supported.
4. Select an IEEE 802.11 packet and expand its header to examine the frame control field, address fields, duration, and sequence control.
5. Compare the frame formats and identify their similarities and differences.

Observations:

Wired Ethernet frames contain source and destination MAC addresses and a type/length field. Wireless LAN frames contain additional control information, and their address fields vary according to the frame type and direction of transmission. Both technologies use MAC addresses and error-detection information to support reliable communication at the data-link layer.



Result:

The comparison shows that wired Ethernet and wireless LANs perform similar data-link-layer functions but use different frame formats. Ethernet is designed for communication over wired links, while IEEE 802.11 includes additional control and addressing information to support wireless communication and mobility.